Delcourt, Michelle; Postle, Luke

Finding an almost perfect matching in a hypergraph avoiding forbidden submatchings.

A *partial* (n,q,r)*-Steiner system*, where $q > r \geq 2$, is a family of q-subsets of an n-element set in which every r-subset belongs to at most one member; in an (n,q,r)-Steiner system, every r-subset belongs to exactly one member. A (j,i)*-configuration* consists of i members whose union has at most j elements. *Erdős* ["Problems and results in combinatorial analysis", Colloq. Int. Teorie Comb., Roma 1973, 3–17 (1976; Zbl 0361.05037)] conjectured that, for every fixed $g \geq 2$, all sufficiently large admissible n admit an $(n,3,2)$-Steiner system without $(i+2,i)$-configurations for $2 \leq i \leq g$. Approximate versions were established by *Glock–Kühn–Lo–Osthus* [Combinatorica 40, No. 3, 363–403 (2020; Zbl 1463.05030)] and independently by *Bohman–Warnke* [J. Lond. Math. Soc. 100, No. 3, 895–913 (2019; Zbl 1433.05050)]; the conjecture was subsequently proved by *Kwan–Sah–Sawhney–Simkin* [Ann. Math. (2) 200, No. 3, 1059–1156 (2024; Zbl 1565.05013)]. Observing that every (n,q,r)-Steiner system contains $(i(q-r)+r+1,i)$-configurations for every fixed $i \geq 2$, Glock–Kühn–Lo–Osthus [loc. cit.] also conjectured its extension to general $q > r \geq 2$, requiring the absence of $(i(q-r)+r,i)$-configurations for $2 \leq i \leq g$. The first main result of the reviewed paper (Theorem 1.3) establishes this extension approximately: for fixed q,r,g, there is a partial (n,q,r)-Steiner system with no such configurations and with at least

$$(1-n^{-\beta})\frac{\binom{n}{r}}{\binom{q}{r}}$$

members, for some $\beta > 0$ and all sufficiently large n.

The underlying result concerns matchings avoiding forbidden submatchings. A *configuration hypergraph* H for a hypergraph G has vertex set $E(G)$, and each edge of H is a matching of G of size at least two. A matching of G is H*-avoiding* if it contains no edge of H. Theorem 1.9 produces an H-avoiding matching covering almost all vertices of a nearly regular uniform hypergraph G, subject to codegree bounds on G, degree and codegree bounds involving H, and a girth condition on H. With H empty, this is a form of Pippenger's almost-perfect-matching theorem, albeit under stronger codegree assumptions; taking G to be a matching, it recovers the mixed-uniformity form of the theorem of *Ajtai–Komlos–Pintz–Spencer–Szemerédi* [J. Comb. Theory, Ser. A 32, No. 3, 321–335 (1982; Zbl 0485.05049)] on independent sets in hypergraphs of girth at least five. The authors frame this union of the two streams as an "edge+vertex phenomenon": the guiding question is when the independence and chromatic numbers of the union $L(G) \cup H$ of the line graph of G with H are close to the respective extremes of its two constituents. The more general Theorem 1.16 allows substantially weaker codegree bounds on G and replaces the girth condition on H with suitable codegree conditions.

The paper also treats bipartite hypergraphs $G = (A,B)$, with every edge meeting A exactly once: Theorem 1.11 gives sufficient conditions for an H-avoiding matching covering all of A, and Theorem 1.12 obtains many pairwise disjoint such matchings. These results also yield coloring and list-coloring bounds for $L(G) \cup H$. Further applications concern approximate high-girth designs, decompositions, and high-dimensional permutations, as well as rainbow matchings. Among the latter is a proof of a conjecture of *Chakraborti–Loh* ["Large rainbow matchings in edge-colored graphs", Preprint, arXiv:2011.04650 (2020)]: for every $\varepsilon > 0$ and all sufficiently large q, a simple bipartite graph properly edge colored with $(1+\varepsilon)q$ colors, each appearing exactly q times, has a rainbow matching of size q.

The principal tool is an iterative version of Rödl's nibble, with an equalizing coin flip so that every edge is deleted on account of forbidden configurations with the same probability, and a weighted-degree parameter to control such configurations through the iterations. Concentration is supplied by a linear-certificate variant of Talagrand's inequality, developed in the paper for this purpose, and the Lovász local lemma completes the iterations. A random sparsification argument leads to the more general theorem.

Related results were obtained independently by *Glock–Joos–Kim–Kühn–Lichev* [J. Lond. Math. Soc. (2) 109, No. 5, Article ID e12899 (2024; Zbl 1539.05108)] using a random greedy process rather than the nibble.

The exact conjecture of Glock–Kühn–Lo–Osthus [loc. cit.], asserting the existence of high-girth (n,q,r)-Steiner systems for all sufficiently large admissible n, remains open in general.

Reviewer: Brahadeesh Sankarnarayanan (Jodhpur)

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